Machine Learning

Outline

- What is Machine Learning?
- Supervised Learning
- Unsupervised Learning
- Overfitting and underfitting
- Regression
- Logistic Regression
- Demo

What is Machine Learning?

- Machine Learning is a branch of Artificial Intelligence (AI) and computer science which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving it's accuracy.

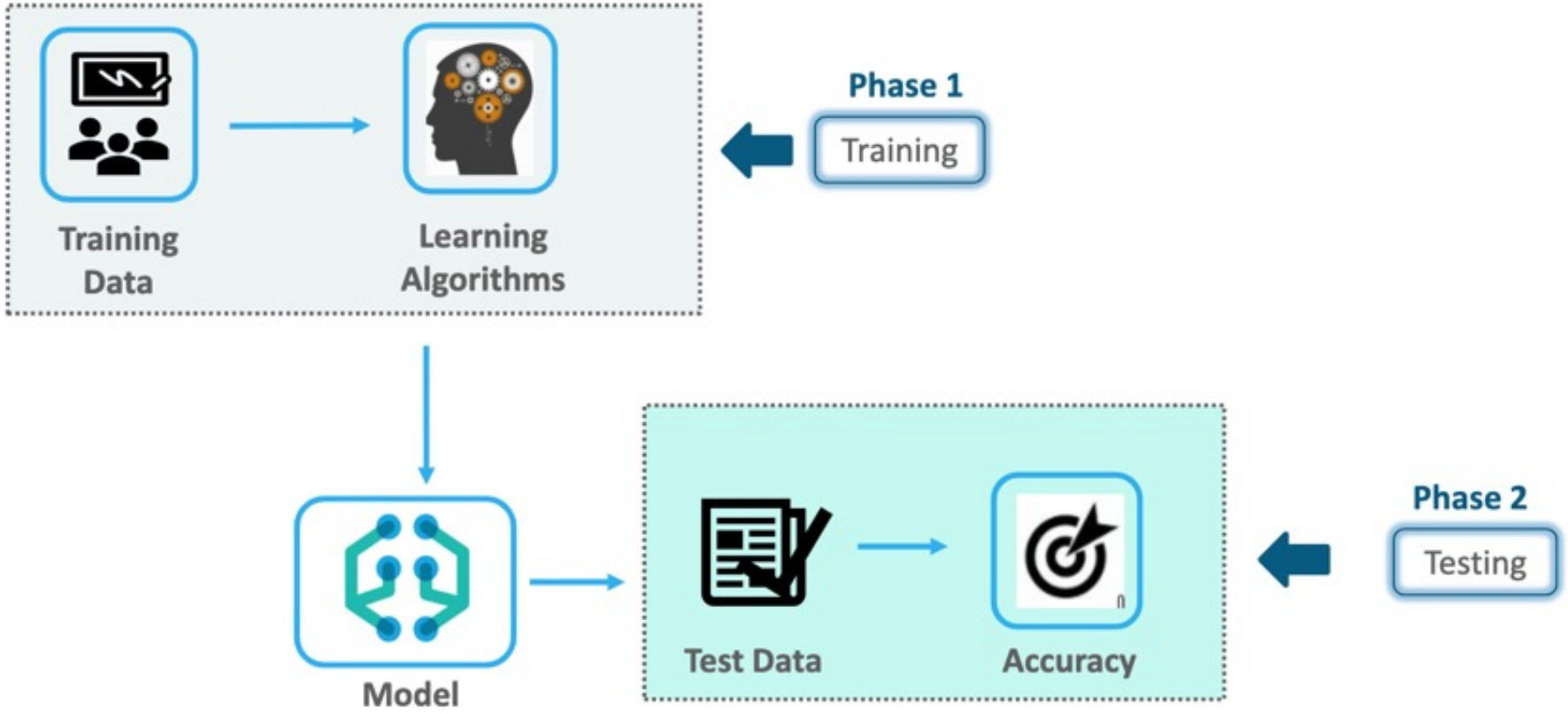
- Machine Learning is an important component of the growing field of data science.

- By using statistical methods, algorithms are trained to make classifications or predictions uncovering key insights.

Applications of Machine Learning

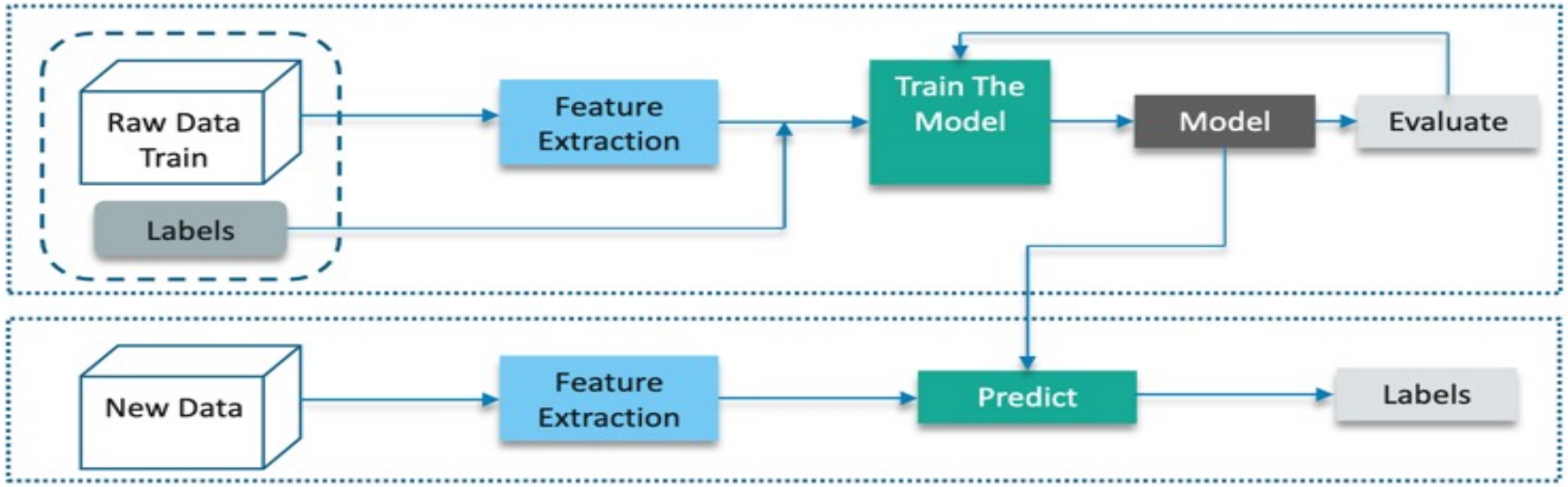
- Siri: Apple claims that the software adapts to user's individual preferences overtime and personalizes results
- Marketing and Sales: The ability to capture data, analyze it, and use it to personalize a shopping experience
- Health Care: Machine Learning involves the use of wearable devices and sensors to assess a patient's heath in real time
- Financial Services: Financial industry use machine learning technology to identify insights in data and prevent fraud

Phases of Machine Learning



Supervised Learning

- Supervised Learning is where you have input variables (X) and an output variable (Y), and you use an algorithm to learn the mapping function from the input to the output.
- Y = F(X)



Twofold cross validation

- In twofold cross-validation, the data are partitioned, using random assignment, into a training data set and a test data set.

- The training data set includes the target variable and the test data does not; the training data set will be pre-classified.

Supervised Learning Algorithms

- Linear Regression: Used to estimate real values based on continuous variable(s)
- Logistic Regression: Used to estimate discrete values based on give set of independent variable(s)
- Decision Tree: Used for classification problems. It works for both categorical and continuous dependent variables
- Random Forest: It's an ensemble of decision trees. It gives better prediction and accuracy than decision tree
- K-Nearest Neighbors: An object is classified by a plurality vote of its neighbors, with the object being assigned to the class most common among its *k* nearest neighbors

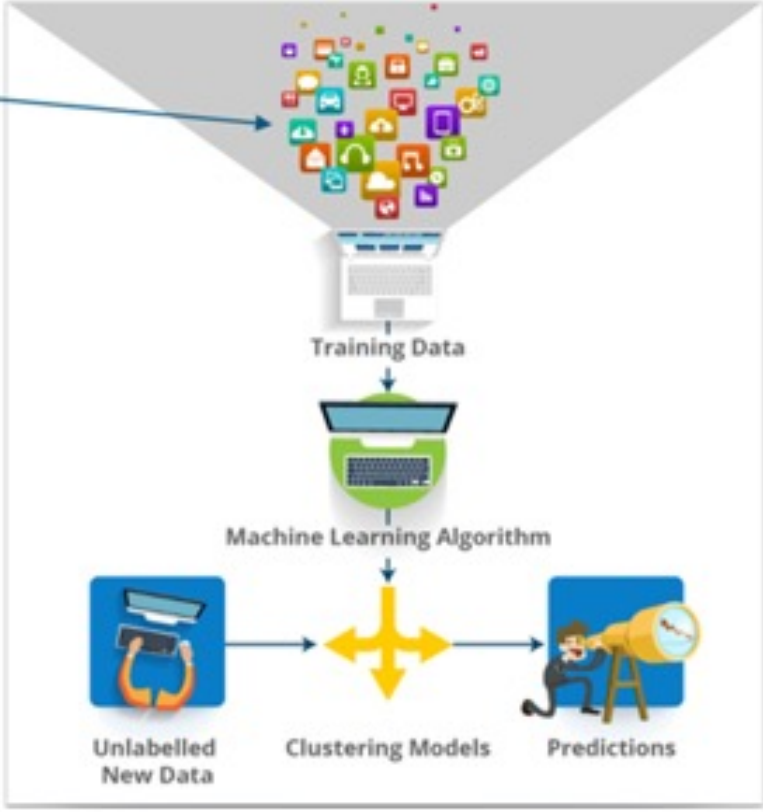
Unsupervised Learning

- Sometime the given data is unstructured and unlabeled. So it becomes difficult to classify that data in different categories

- Unsupervised learning helps to solve this problem. This learning is used to cluster the input data in classes on the basis of their statistical properties

Unsupervised Learning – Process Flow

Training data is collection of information without any label

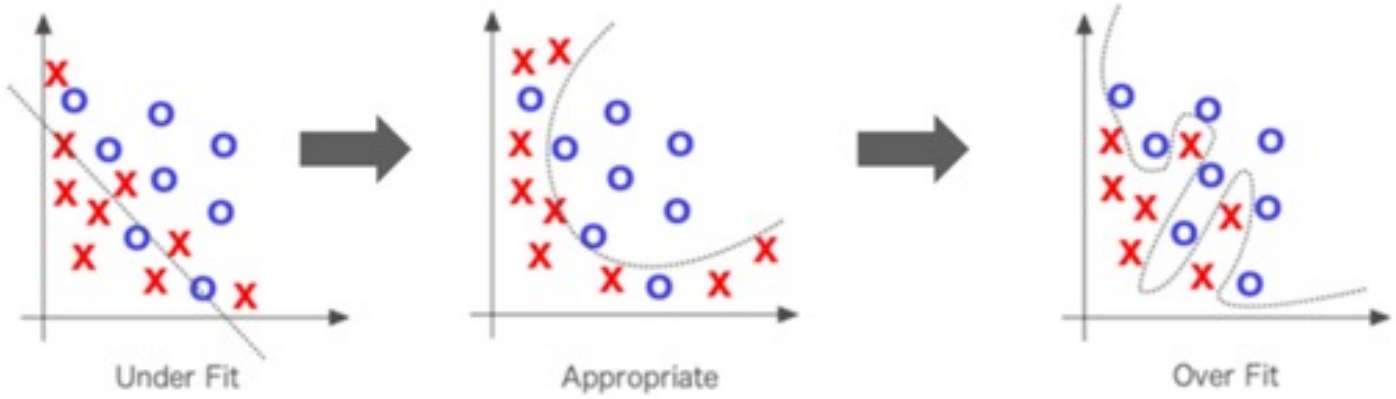


Unsupervised Learning Algorithms

- K-Means Clustering
- Hierarchical Clustering

Types of Model Fitting: Underfitting And Overfitting

- ML Algorithms first attempt to solve the problem of under-fitting; that is, of taking a line that does not approximate the data well, and making it to approximate the data better
- ML doesn't know where to stop to solve the problem it can even go ahead from Appropriate to Over Fit model. When we say a model overfits a dataset we mean, it may have a low error rate for the training data, but it may not generalize well to overall population of data we're interested in



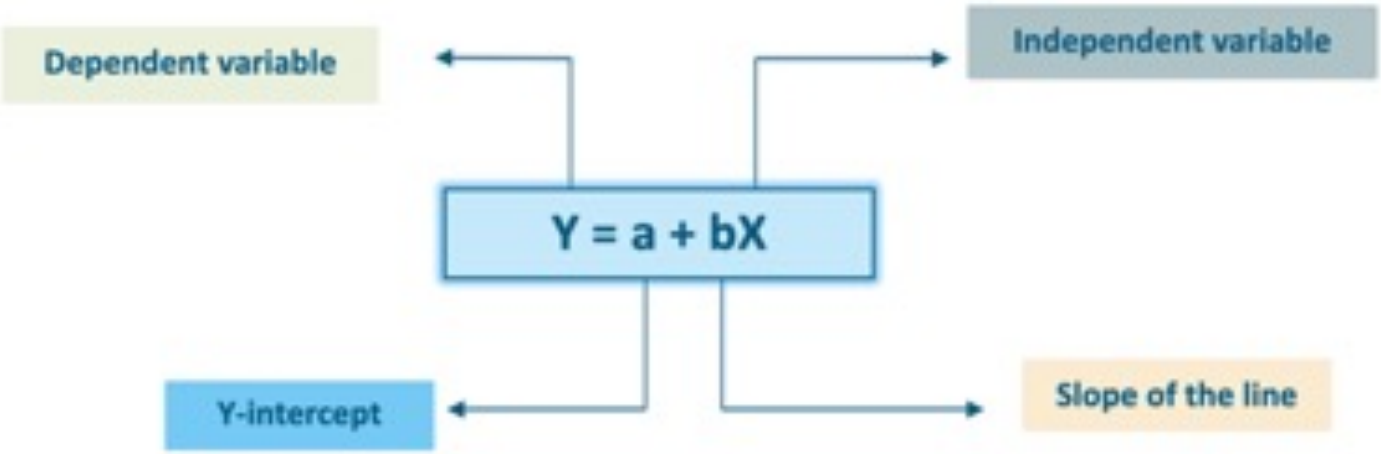
Linear Regression

- Linear Regression Analysis is a powerful technique used for predicting unknown value of a variable (Dependent Variable) from the known value of another variables (Independent Variable)

- A Dependent Variable (DV) is the variable to be predicted or explained in a regression model

- An Independent Variable (IDV) is the variable related to the dependent variable in a regression equation

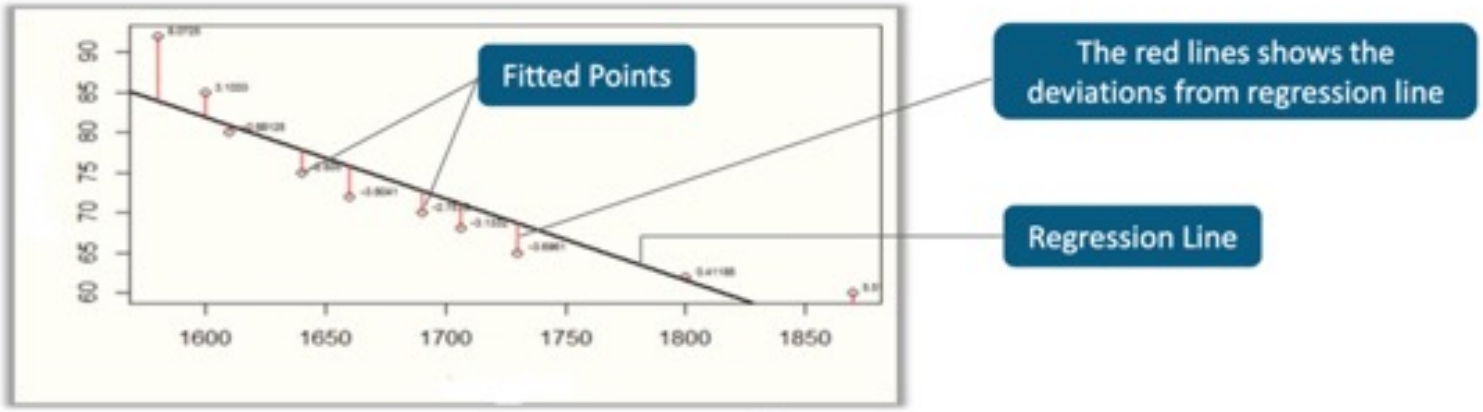
Simple Linear Regression



The Regression Line

- The regression is simply a single line that best fits the data (in terms of having the smallest overall distance from the line to the points)

- This technique is used for finding the "best-fitting-line" using the "Least Squares Method"



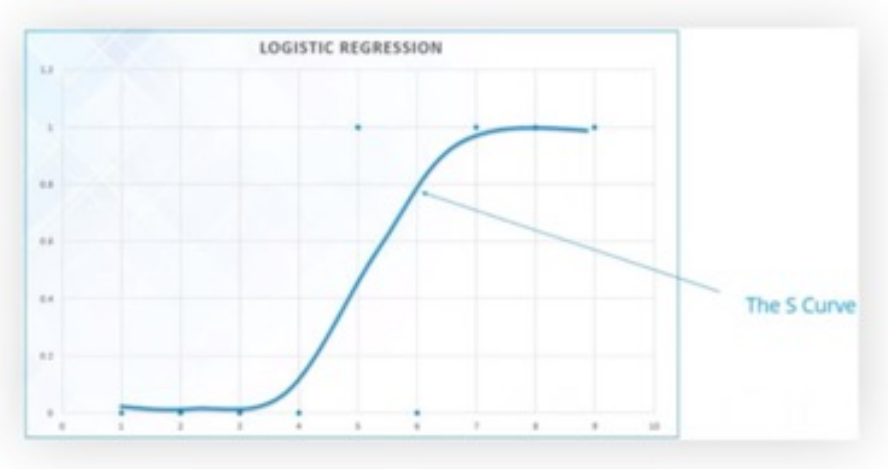
Logistic Regression

- Logistic Regression is a statistical method for analyzing a data set in which there are one or more independent variables that determine an outcome - Outcome is a binary class type.

- The logistic regression curve is called as "Sigmoid Curve".

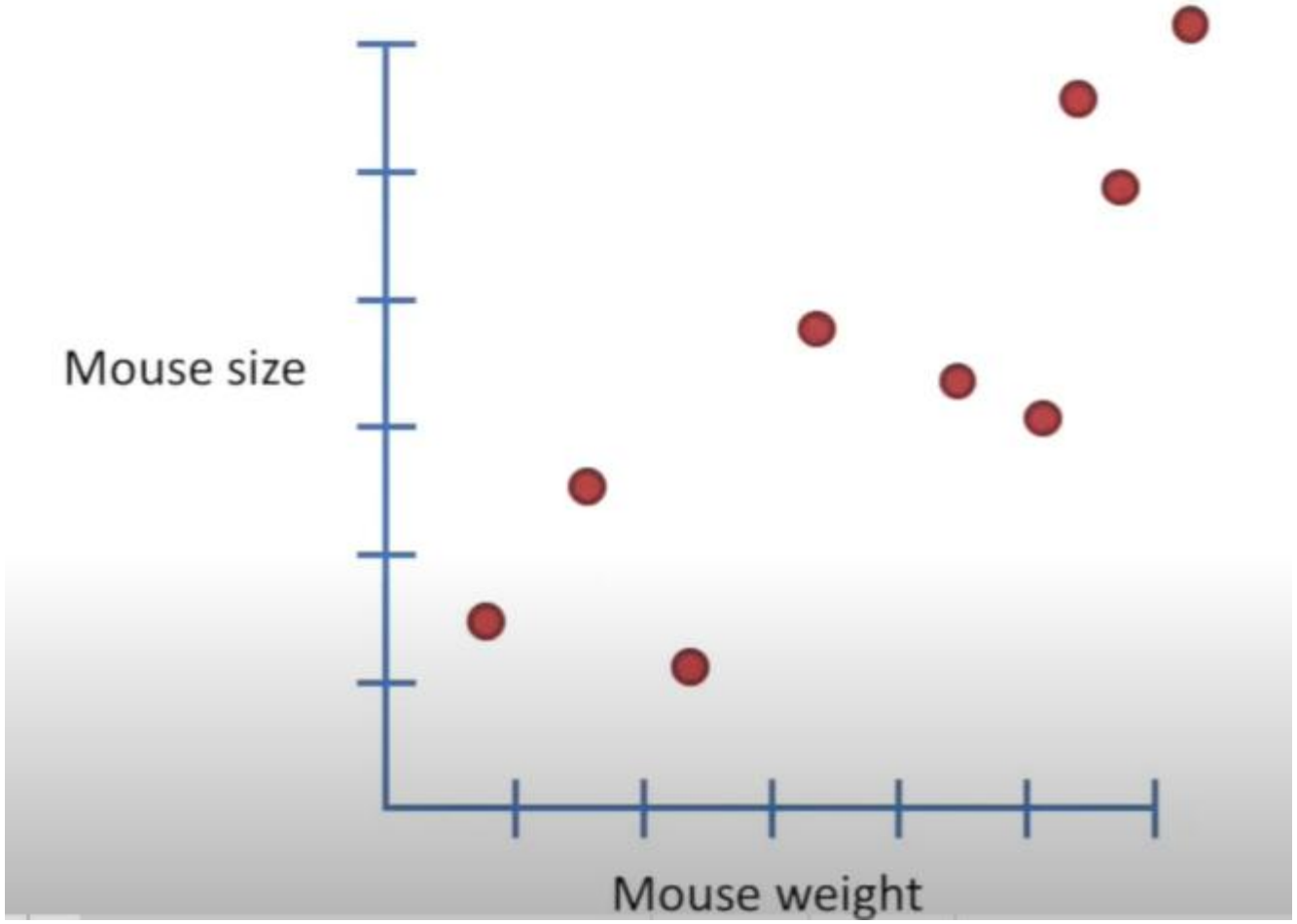
- The sigmoid function converts any value from

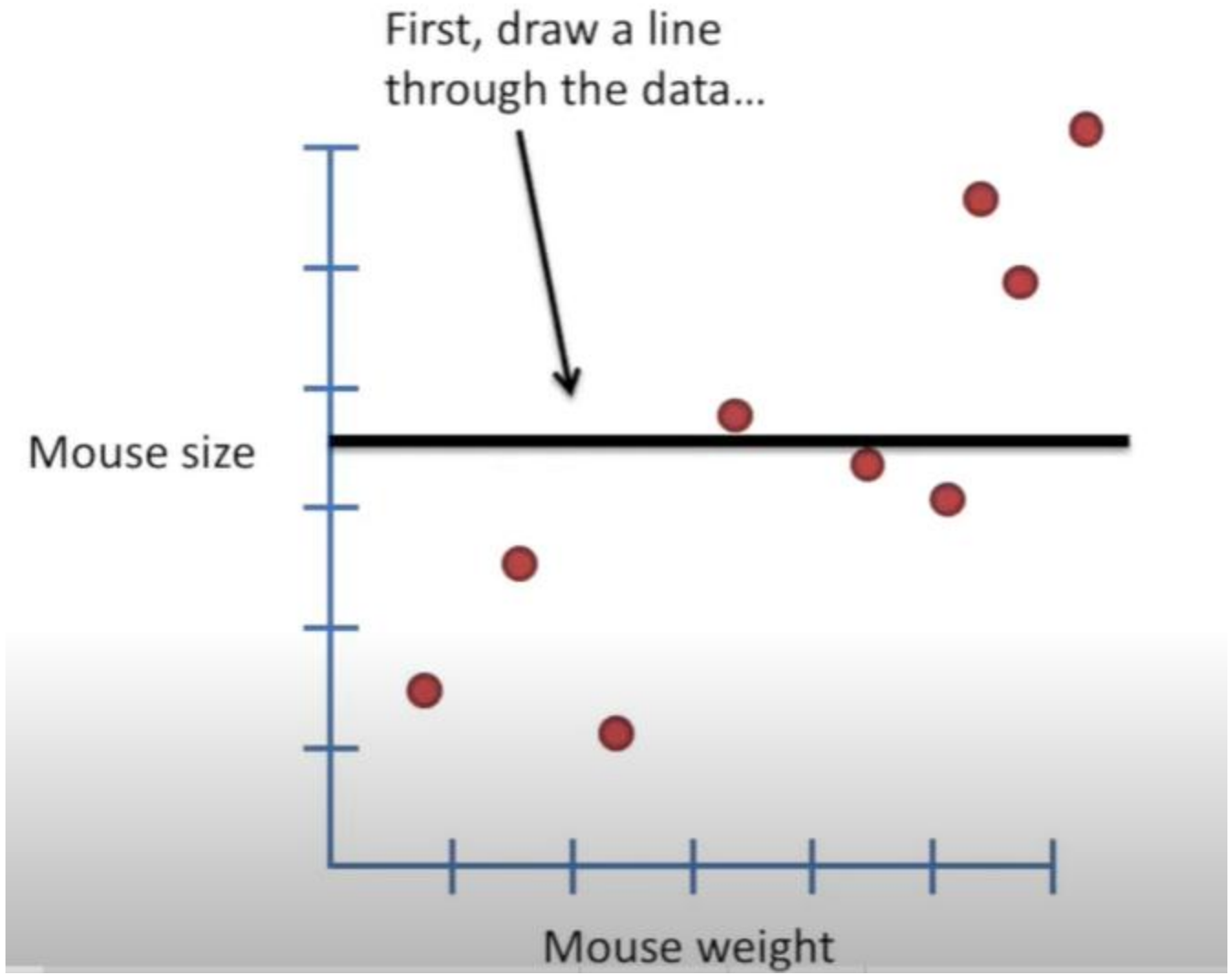
  -infinity to infinity to the discrete values 0 to 1

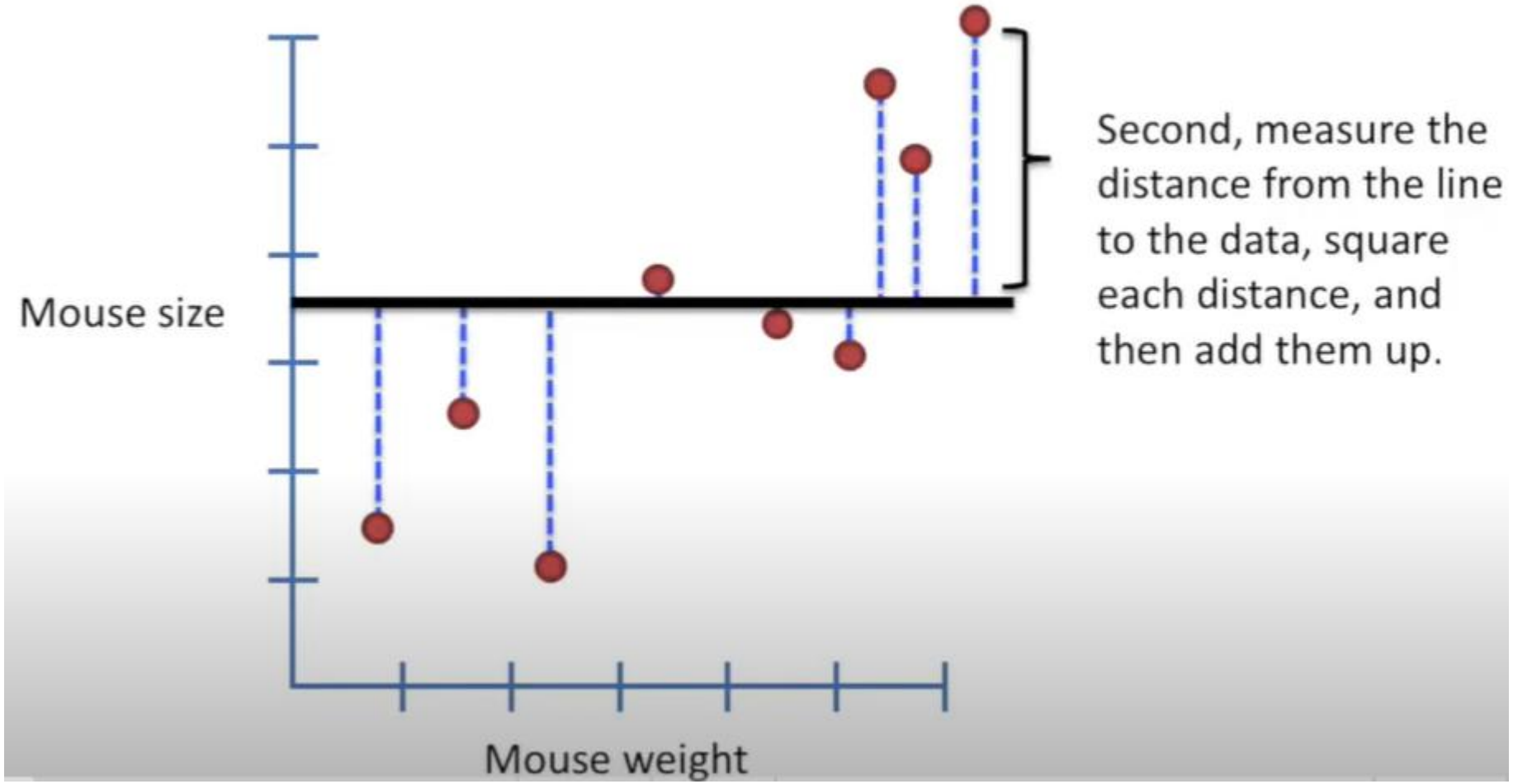


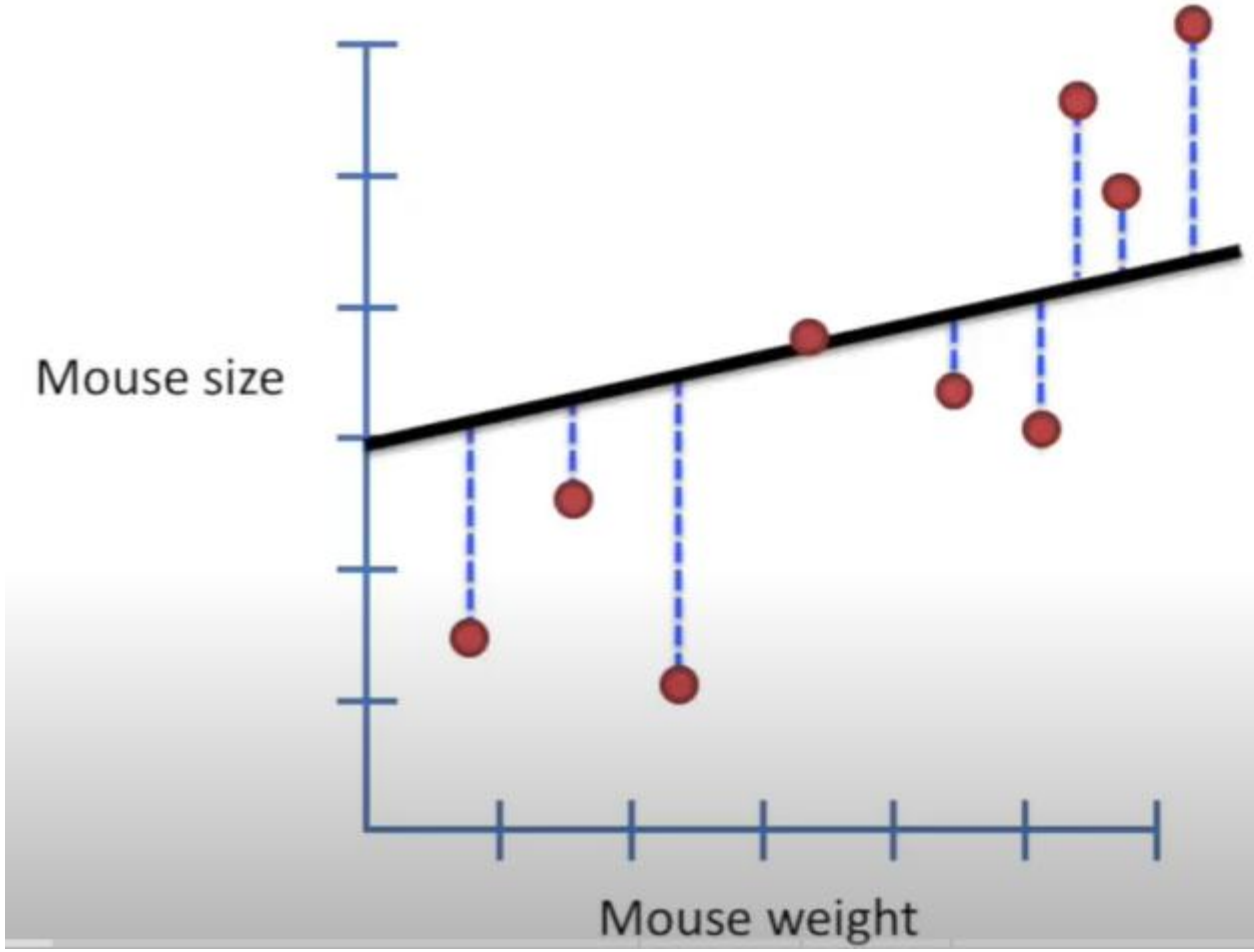
$$\log\left[\frac{Y}{1-Y}\right] \Rightarrow Y = C + B1X1 + B2X2 + \ldots \longrightarrow \text{Final Logistic Regression Equation}$$

Linear Regression Example

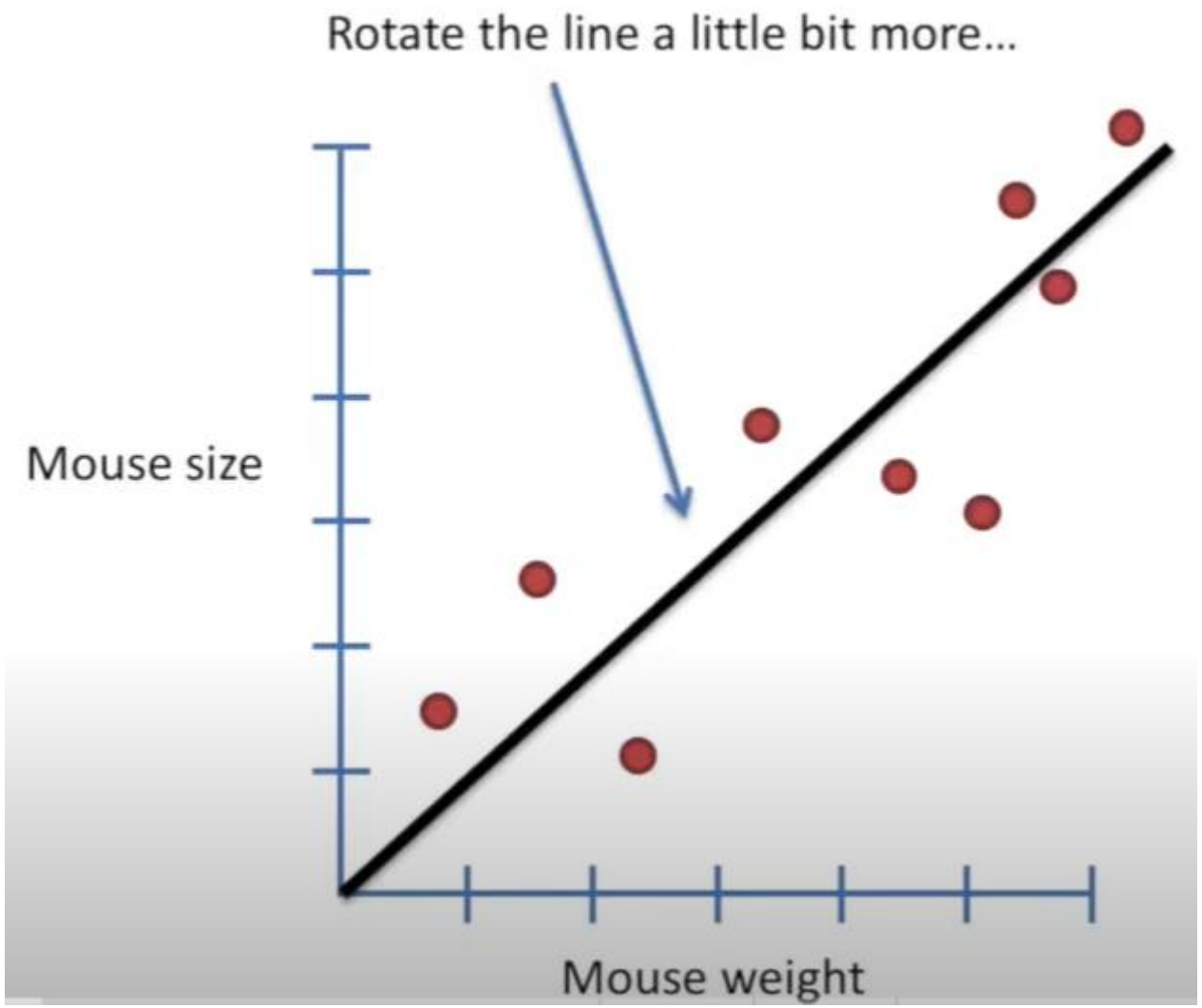

Mouse size
Mouse weight

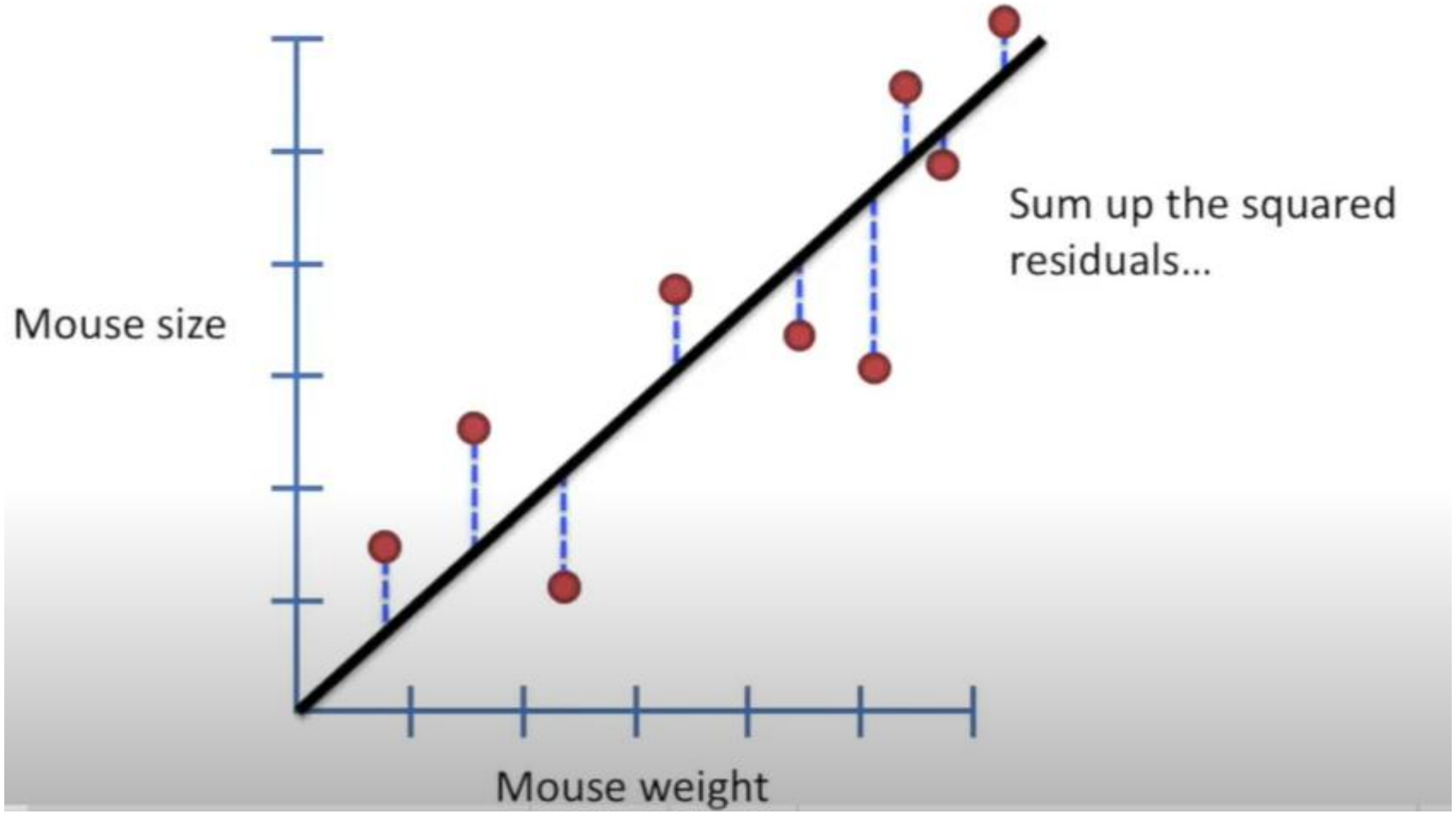

First, draw a line
through the data…
Mouse size
Mouse weight

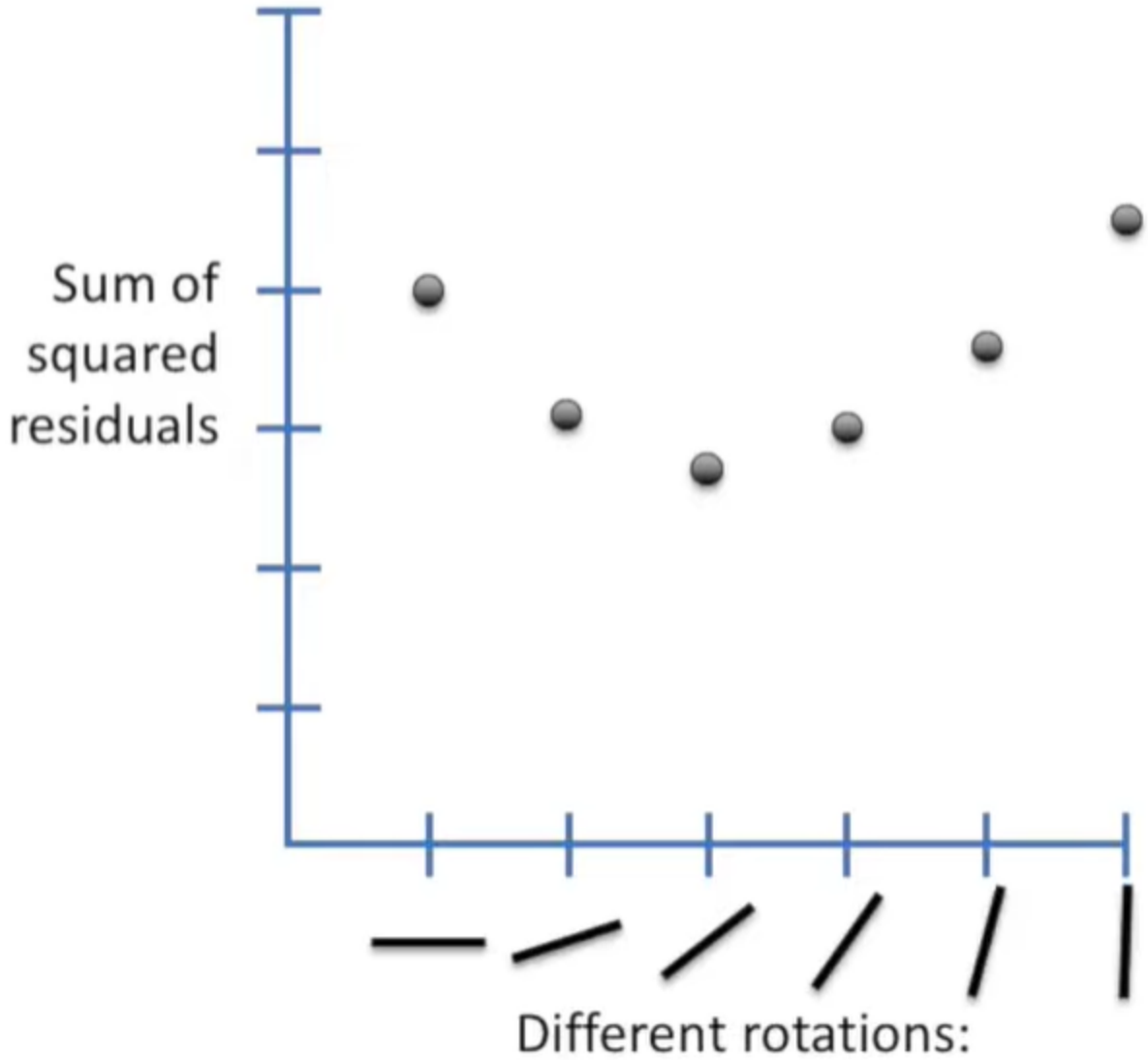

Mouse size
Mouse weight
Second, measure the distance from the line to the data, square each distance, and then add them up.

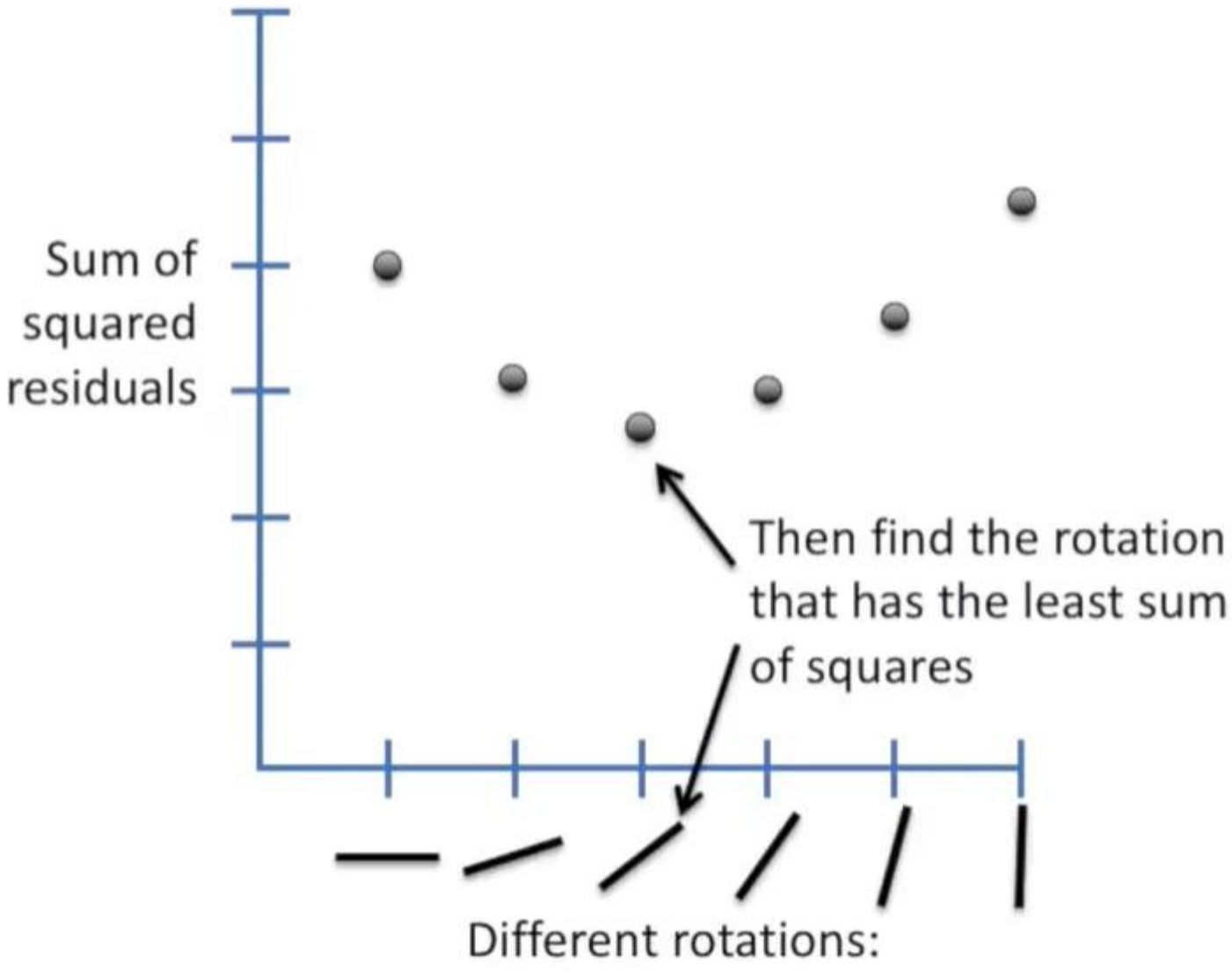


With the new line, measure the residuals, square them, and then sum up the squares.


Rotate the line a little bit more…
Mouse size
Mouse weight


Sum up the squared residuals…
Mouse size
Mouse weight


Sum of squared residuals
Different rotations:

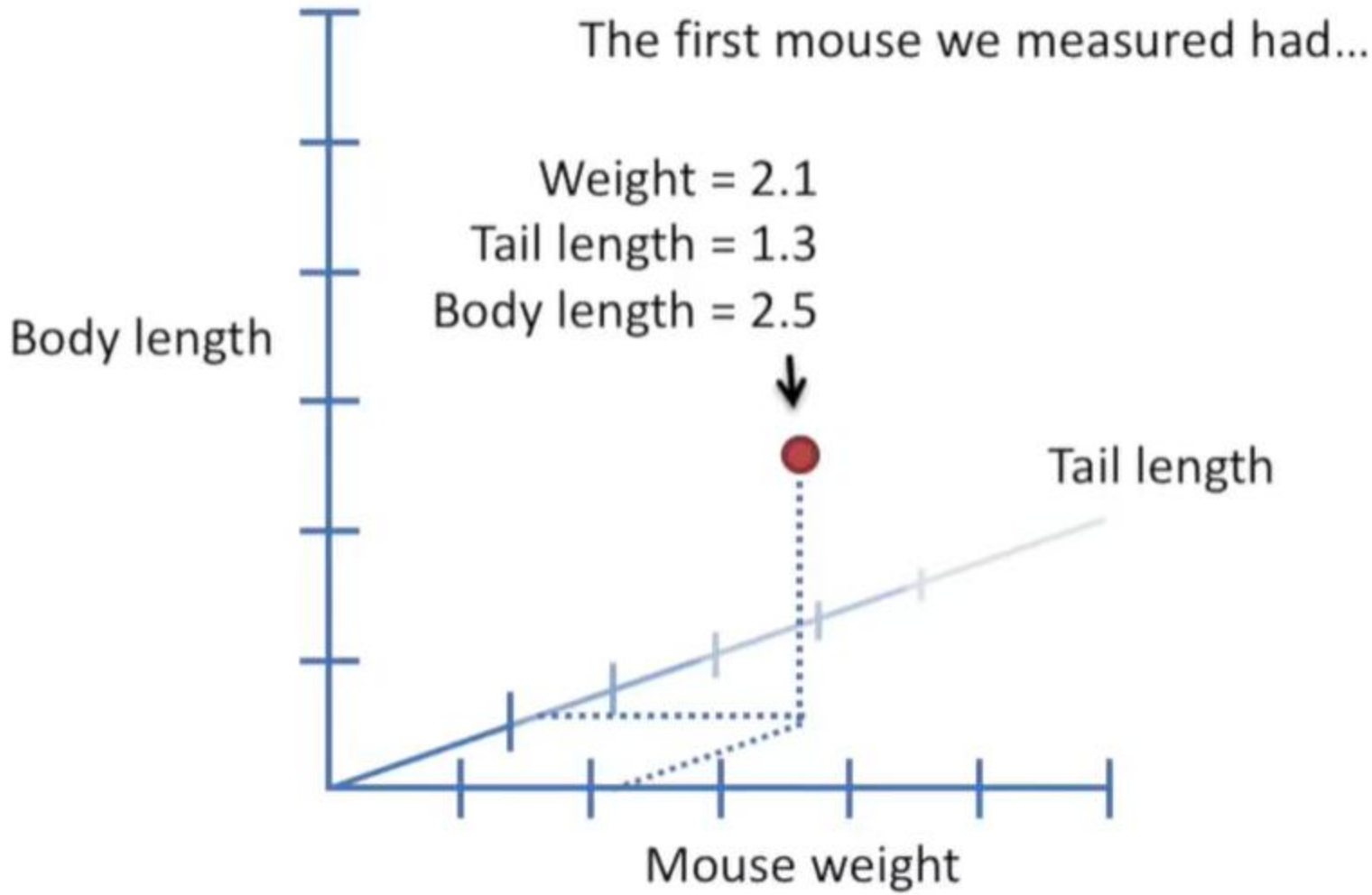

Sum of squared residuals
Then find the rotation that has the least sum of squares
Different rotations:

Let's look at slightly more complicated example:

Imagine we wanted to know if mouse **weight** and **tail length** did a good job predicting the **length of the mouse's body**.

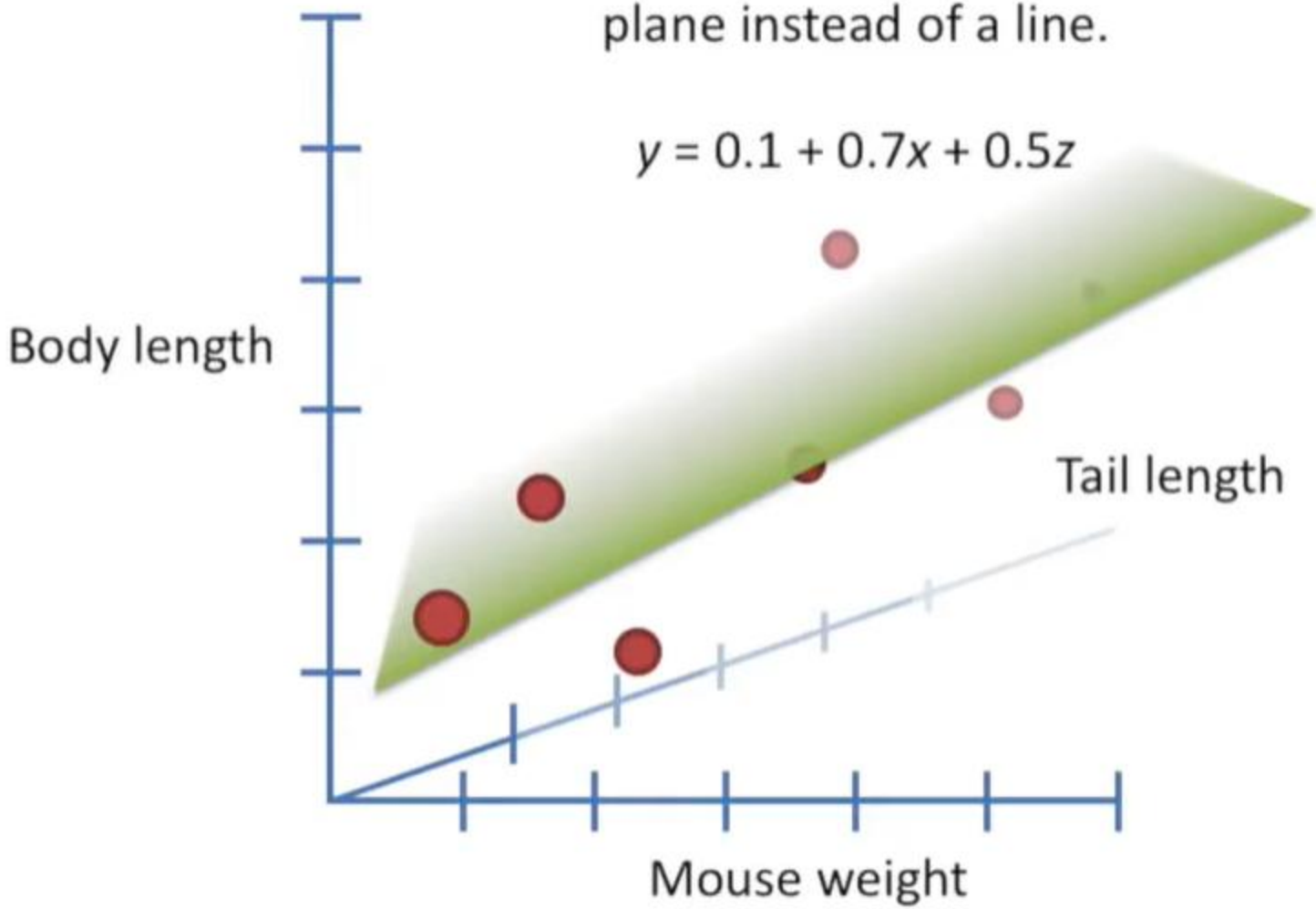
So we measured a bunch of mice…

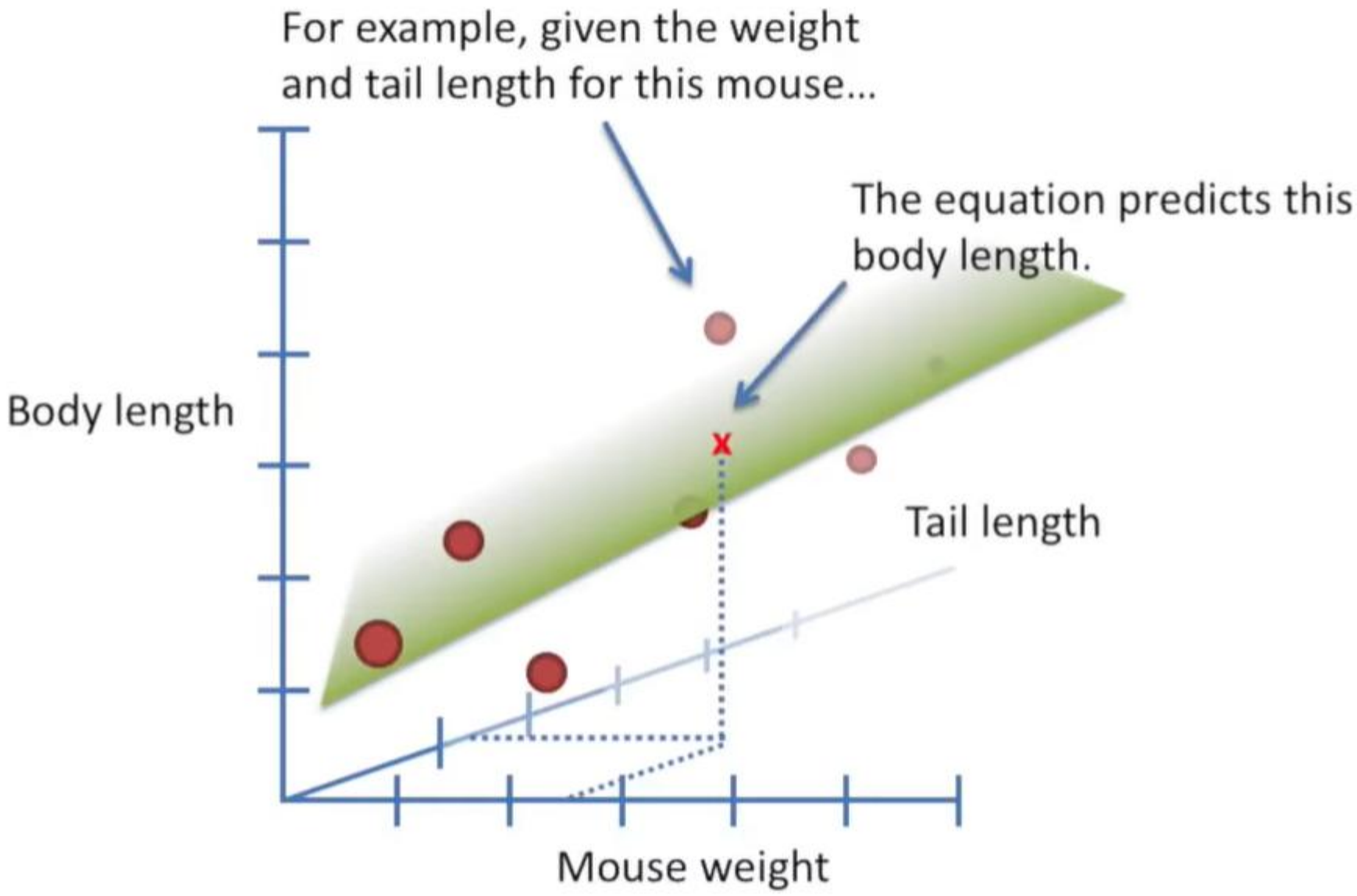
Weight	Tail Length	Body Length
3.5	2.9	3.1
1.3	2.1	2.8
5.9	4.1	6.1
4.8	3.2	3.8
…	…	…


The first mouse we measured had…
Weight = 2.1
Tail length = 1.3
Body length = 2.5
Body length
Tail length
Mouse weight

Now we do a least-squares fit. Since we have the extra term in the equation, we fit a plane instead of a line.

$$y = 0.1 + 0.7x + 0.5z$$




For example, given the weight and tail length for this mouse…
The equation predicts this body length.
Body length
Tail length
Mouse weight